

PostgreSQL pg_dump - Detailed Backup Guide

1. Plain SQL Backup (-Fp)

Purpose: Creates a readable SQL script containing schema and data.

When to use: Small/medium databases, migrations, learning, or when you want to inspect the SQL.

Restore: Use psql.

```
Backup:  
pg_dump -Fp -d companydb -f /var/lib/pgsql/16/pgbackup/companydb.sql
```

```
Restore:  
psql -d companydb -f /var/lib/pgsql/16/pgbackup/companydb.sql
```

2. Custom Backup (-Fc)

Purpose: Creates a PostgreSQL custom-format backup.

When to use: Production backups, selective restore, compressed backups.

Restore: Use pg_restore.

```
Backup:  
pg_dump -Fc -d companydb -f /var/lib/pgsql/16/pgbackup/companydb.backup
```

```
Restore:  
pg_restore -d companydb /var/lib/pgsql/16/pgbackup/companydb.backup
```

3. Tar Backup (-Ft)

Purpose: Creates a TAR archive backup.

When to use: Single archive file for transport or storage.

Restore: Use pg_restore.

```
Backup:  
pg_dump -Ft -d companydb -f /var/lib/pgsql/16/pgbackup/companydb.tar
```

```
Restore:  
pg_restore -d companydb /var/lib/pgsql/16/pgbackup/companydb.tar
```

4. Directory Backup (-Fd)

Purpose: Stores backup as multiple files in a directory.

When to use: Large databases and parallel backup/restore.

Restore: Use pg_restore.

```
Backup:  
pg_dump -Fd -d companydb -f /var/lib/pgsql/16/pgbackup/companydb_dir
```

```
Restore:  
pg_restore -d companydb /var/lib/pgsql/16/pgbackup/companydb_dir
```

5. Single Table (-t)

Purpose: Backup one table only.

When to use: Restore or migrate a specific table.

```
Backup:  
pg_dump -d companydb -t company.employees -f /var/lib/pgsql/16/pgbackup/employees.sql
```

6. Multiple Tables (-t)

Purpose: Backup only selected tables.

When to use: Application-specific or partial backups.

```
pg_dump -d companydb -t company.departments -t company.employees -t company.customers -f /var/lib/pg
```

7. Schema Only (-s)

Purpose: Backup database objects without data.

When to use: Deploy schema to another environment.

```
pg_dump -s -d companydb -f /var/lib/pgsql/16/pgbackup/company_schema.sql
```

8. Data Only (-a)

Purpose: Backup only table data.

When to use: Load data into an existing schema.

```
pg_dump -a -d companydb -f /var/lib/pgsql/16/pgbackup/company_data.sql
```

9. Single Schema (-n)

Purpose: Backup one schema with its objects and data.

When to use: Databases containing multiple schemas.

```
pg_dump -d companydb -n company -f /var/lib/pgsql/16/pgbackup/company_schema_backup.sql
```

10. Multiple Schemas (-n)

Purpose: Backup selected schemas.

When to use: Migrate or restore only required schemas.

```
pg_dump -d companydb -n company -n hr -n sales -f /var/lib/pgsql/16/pgbackup/multiple_schemas.sql
```

11. Exclude Table (-T)

Purpose: Exclude specific tables.

When to use: Skip audit or temporary tables.

```
pg_dump -d companydb -T company.audit_log -f /var/lib/pgsql/16/pgbackup/company_without_audit.sql
```

12. Verbose (-v)

Purpose: Display backup progress.

When to use: Monitoring long-running backups.

```
pg_dump -v -d companydb -f /var/lib/pgsql/16/pgbackup/company_verbose.sql
```

13. Compressed Backup (-Z9)

Purpose: Create a compressed custom backup.

When to use: Save storage space.

```
pg_dump -Fc -Z9 -d companydb -f /var/lib/pgsql/16/pgbackup/companydb_compressed.backup
```

14. Parallel Backup (-Fd -j2)

Purpose: Run backup using multiple worker processes.

When to use: Large databases. Works only with directory format.

Backup:

```
pg_dump -Fd -j2 -d companydb -f /var/lib/pgsql/16/pgbackup/companydb_parallel
```

Restore:

```
pg_restore -j2 -d companydb /var/lib/pgsql/16/pgbackup/companydb_parallel
```